భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad
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Matrix Theory

20 Eigenvalues and Eigenvectors - I

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Eigenvectors

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- Determinants give this transition
- All our matrices are square now

A matrix usually turns a vector

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Apply it to a few vectors:

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \qquad A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}, \qquad A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix}.$$

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The question for the next three classes

Are there directions that A does *not* turn?

Two special directions

For the same $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$:

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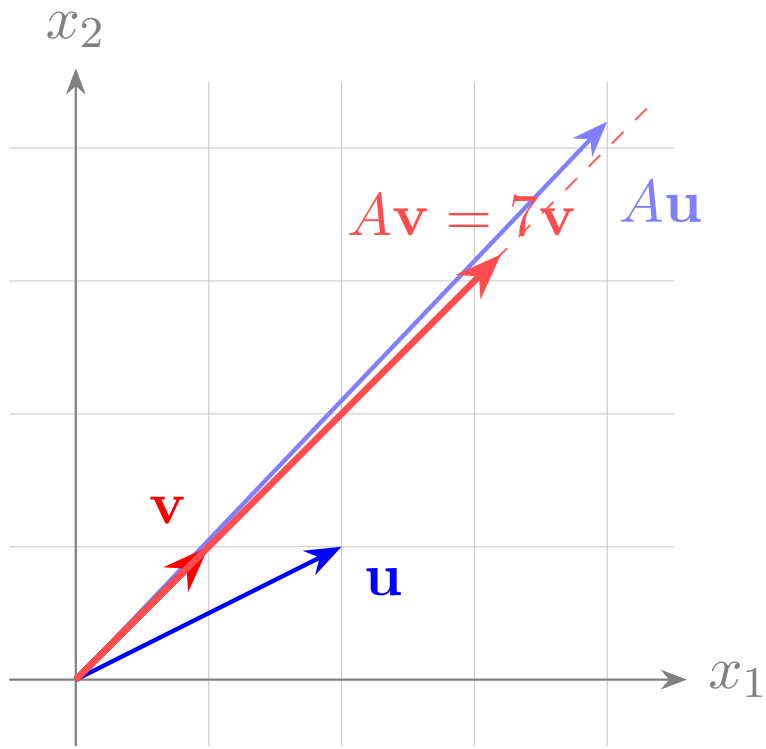
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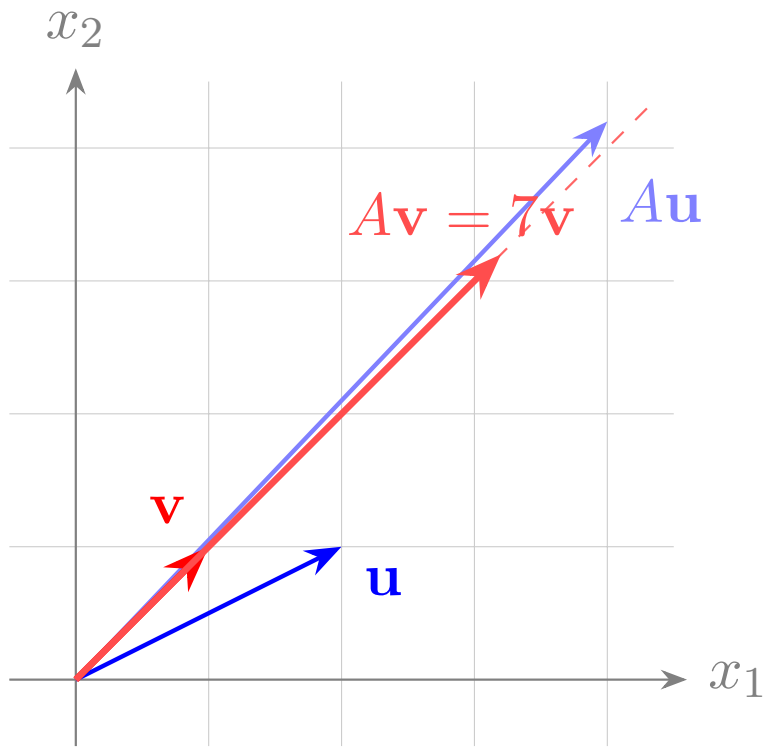
Why we care

On such a direction, the complicated object A collapses to scalar multiplication. If we can decompose a problem into these directions, linear algebra becomes arithmetic.

Example Visualization



Example Visualization



$\mathbf{u}$ leaves its line; $\mathbf{v}$ stays on its line. The dashed line is *invariant* under A.

Definition

Eigenvector and eigenvalue

Let A be an $n \times n$ matrix. A **nonzero** vector $\mathbf{v} \in \mathbb{R}^n$ is an *eigenvector* of A if

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some scalar λ. That scalar λ is the *eigenvalue* of A corresponding to $\mathbf{v}$.

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- $\mathbf{v} \neq \mathbf{0}$ is part of the definition. ($A\mathbf{0} = \lambda\mathbf{0}$ always holds and says nothing.)
- $\lambda = 0$ *is* allowed. $A\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$ means 0 is an eigenvalue.

Verifying a vector

Question

Is $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ an eigenvector of $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$?

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Note the asymmetry

Verifying a given vector is a one-line computation. *Finding* the eigenvectors is the real work.

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We need a nonzero $\mathbf{v}$ with $A\mathbf{v} = 7\mathbf{v}$, i.e.

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$$A - 7I = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} \xrightarrow{\text{row reduce}} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

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$$A - 7I = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} \xrightarrow{\text{row reduce}} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

A free variable exists, so there are nonzero solutions: $\mathbf{v} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Yes, 7 is an eigenvalue.

Eigenspace

For an eigenvalue λ of A, the *eigenspace* of A for λ is

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- λ is an eigenvalue $\iff$ $\mathrm{Nul}(A - \lambda I) \neq \{\mathbf{0}\}$ $\iff$ $A - \lambda I$ is **not invertible**.

Eigenspace example

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Every vector in that plane is simply doubled by A.

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Characteristic equation

$$\det(A-\lambda I) = 0$$

is the *characteristic equation* of A; the left side is the *characteristic polynomial* $p(\lambda)$.

The characteristic polynomial

Theorem

If A is $n \times n$, then $p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n in λ, with leading term $(-1)^n \lambda^n$.

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- Counted with repetition, there are exactly n.
- Over $\mathbb{R}$ there may be **none** .

Algebraic Multiplicity

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The *algebraic multiplicity* of λ_i is its multiplicity as a root of $p(\lambda)$.

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Expanding:

$$p(\lambda) = \lambda^2 - (a+d)\lambda + (ad - bc) = \lambda^2 - \mathrm{tr}(A)\,\lambda + \det(A).$$

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Worth memorizing

For 2×2 matrices: $\lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0$.

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$\lambda = 5$**:** $A - 5I = \begin{bmatrix} 2 & 2 \\ -4 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, so $x_1 = -x_2$ and $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

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$\lambda = 3$: $A - 3I = \begin{bmatrix} 4 & 2 \\ -4 & -2 \end{bmatrix} \to \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$, so $x_2 = -2x_1$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

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Check: $A \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 - 4 \\ -4 - 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. ✓

3×3 Example

$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. Expanding $\det(A - \lambda I)$ gives

$$p(\lambda) = -(\lambda - 2)^2(\lambda - 9).$$

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For $\lambda = 9$:

$$A - 9I = \begin{bmatrix} -5 & -1 & 6 \\ 2 & -8 & 6 \\ 2 & -1 & -1 \end{bmatrix} \longrightarrow \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Check: $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 9 \\ 9 \end{bmatrix}$. ✓

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Check: $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 9 \\ 9 \end{bmatrix}$. ✓ Check the trace: $4 + 1 + 8 = 13 = 2 + 2 + 9$. ✓

Note on the computation required

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- Expanding $\det(A - \lambda I)$ costs $\approx n!$ operations
- Polynomial root-finding can be cumbersome and numerically unstable
- In practice, softwares use iterative methods (e.g., QR algorithm) that avoid characteristic polynomial

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theorem

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$A - \lambda I$ is triangular, and the determinant of a triangular matrix is the product of its diagonal entries, so $p(\lambda) = \prod_i (a_{ii} - \lambda)$.

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Example

$$A = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$ has eigenvalues $3,\ 0,\ 2$.

Since 0 is an eigenvalue, A is not invertible.

Zero as an eigenvalue

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This attaches eigenvalues to the Invertible Matrix Theorem: add to the list of equivalent statements

- $\det A \neq 0$;
- 0 is not an eigenvalue of A;
- $\lambda_1 \lambda_2 \cdots \lambda_n \neq 0$.

Invertible Matrix Theorem

Eigenvalues of related matrices

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- $A^k\mathbf{v} = \lambda^k\mathbf{v}$ (apply A repeatedly)

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Not true in general: the eigenvalues of $A + B$ are *not* the sums of those of A and B.

Distinct eigenvalues $\rightarrow$ independent eigenvectors

If $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are eigenvectors of A corresponding to **distinct** eigenvalues $\lambda_1, \ldots, \lambda_r$, then $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly independent.

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Proof sketch (contradiction)

Let p be least with $\mathbf{v}_{p+1} \in \mathrm{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$, say

$$\mathbf{v}_{p+1} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p. \qquad (*)$$

Apply A: $\lambda_{p+1}\mathbf{v}_{p+1} = c_1\lambda_1\mathbf{v}_1 + \cdots + c_p\lambda_p\mathbf{v}_p$.
Multiply $(*)$ by λ_{p+1} and subtract:

$$\mathbf{0} = c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p.$$

By minimality of p the $\mathbf{v}_i$ here are independent, so every $c_i(\lambda_i - \lambda_{p+1}) = 0$. The eigenvalues are distinct, so all $c_i = 0$, forcing $\mathbf{v}_{p+1} = \mathbf{0}$ — impossible.

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- Distinct eigenvalues $\Rightarrow$ independent eigenvectors.